MDS Machine Portal · Trafag AG

ScadaViewer

User Manual

Web application for viewing production and test data of microswitches — Trafag AG line

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1 Introduction

1.1 What the application is for

ScadaViewer is a web application for viewing data from the microswitch measuring and sorting line. During production and tests, the **DatabaseGateway** service writes the measured values of every piece to CSV files on the machine PC and uploads them to network storage (Synology NAS). ScadaViewer presents this data clearly — in tables, charts and diagrams — and lets you export and print it.

In the application you can:

- browse orders (Production) and tests (Testing) on the machine PC and on the NAS,
- follow the order that is currently being measured,
- see how the pieces of an order are distributed into categories 1–6 and how production progressed over time,
- filter the pieces of an order by OK / NOK status and by category,
- open the detail of a single piece with the force–travel diagram and contact switching times,
- view the signal data of tests (position, force, contact voltage, current and resistance),
- download data as CSV or XLSX and print A4 reports,
- monitor how full the local storage is and free up space after synchronisation to the NAS.

> **Tip:** The application only reads data. The only change it can make is deleting **local** copies on the machine PC. ScadaViewer never deletes or modifies files on the NAS.

1.2 Where the application runs and how to open it

The application runs as a service on the **machine PC**. It is always open on the machine monitor (left screen; the machine HMI runs on the right screen). From the office, open it in a browser (Chrome or Edge) at the address provided by your administrator, for example:

```
http://<machine PC address>:8080
```

From the office the application is **view-only**: you can browse, export and print data, but deleting files, cleaning up storage and changing settings is only possible on the machine PC. During remote access the top bar shows the label **Remote access · view only**.

1.3 User roles

Every user has one of four roles. A higher role can do everything a lower one can. The table shows what each role may do:

Table 1 – Permissions by role (Office column = access from a computer other than the machine PC)

Action	Operator	Technician	Administrator	Manufacturer	Office
Viewing, CSV / XLSX export, printing	✓	✓	✓	✓	✓
Clean up synced files	✓	✓	✓	✓	—
Deleting files (incl. batch)	—	✓	✓	✓	—
Storage paths and limit settings	—	—	✓	✓	—

Action	Operator	Technician	Administrator	Manufacturer	Office
User management	—	—	lower roles only	✓	—
Changing your own password	✓	✓	✓	✓	✓

1.4 About this manual

The figures in this manual were taken with sample data. The PLC is disconnected in the figures (red status in the top bar); in operation at the machine the status is green. Order names, times and values are examples only.

2 Sign-in and basic controls

2.1 Signing in

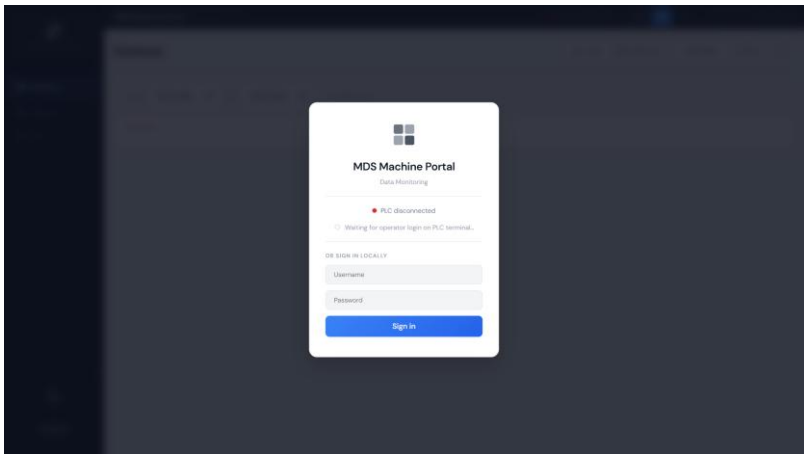


Figure 1 – Sign-in screen

There are two ways to sign in:

- **Automatically via the PLC** — when the operator signs in on the machine terminal, the application on the machine PC signs in by itself. Meanwhile the screen shows "Waiting for operator login on the PLC terminal…". Automatic sign-in does not work from the office.
- **With user name and password** — fill in the fields under "Or sign in locally" and press **Sign in**.

A sign-in is valid for 8 hours. After that the application shows "Session expired — please sign in again." After five wrong password attempts, sign-in from that computer is blocked for 10 minutes. Sign out with the arrow button in the top bar.

2.2 Top bar

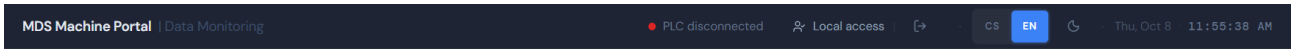


Figure 2 – Top bar

Table 2 – Top bar items

Item	Meaning
PLC connected / disconnected	Connection to the machine controller (green = OK, red = no connection).
User	Who is signed in, or how they signed in.
Exit arrow	Sign out.
CS / EN	Application language. The choice is remembered in the browser.
Moon / sun	Dark or light appearance.
Date and time	Current time of the PC running the application.
Local storage (only when filling up)	Appears when the storage is more than 80 % full. Click it to go to the Database.
Remote access · view only	The application is open on a computer other than the machine PC.

2.3 Menu and controls

The left menu contains the **Database**, **Settings** and **Info** pages. Click the logo to return to the Database. The application is designed for a touch monitor; every button can be operated with a finger.

- **F5** — refreshes the file list in the Database (without reloading the whole page).
- **Esc** — closes an expanded order or an open dialog.
- If the connection to the application server is lost, a red bar “Server unavailable — checking connection…” appears. It disappears by itself once the connection is restored.

3 Database

The Database is the main page of the application. It lists files — one file is one order (Production) or one test (Testing).

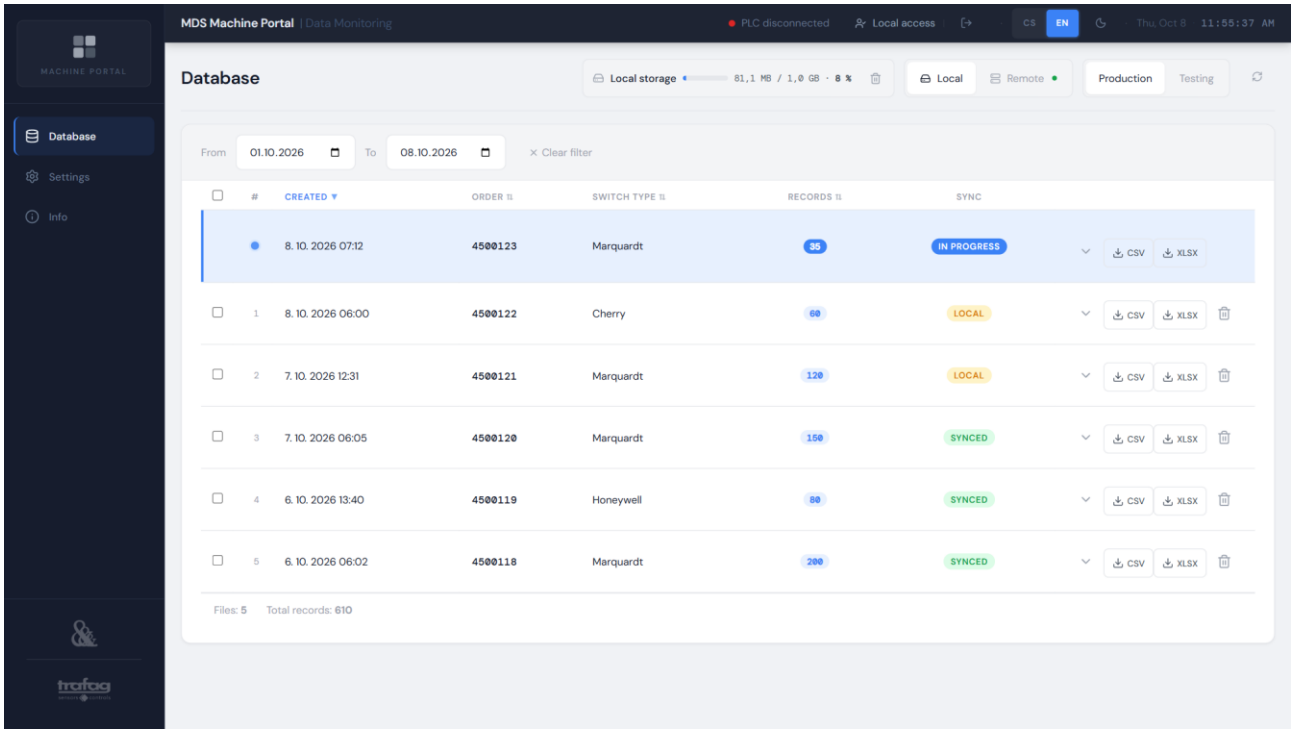


Figure 3 – Database — orders on the machine PC

3.1 Order list

Table 3 – List columns

Column	Meaning
Created	When measuring of the order started.
Order	Order number.
Switch type	Microswitch type (manufacturer).
Records	Number of measured pieces (“…” = still being counted).
Sync	Synchronisation status to the NAS — see the next table.
Buttons on the right	Expand the order, download CSV, download XLSX, delete.

Table 4 – Synchronisation status (Sync column)

Badge	Meaning
In progress	The order is being measured right now. It is always at the top and its piece count updates automatically. It cannot be deleted.
Local	The order is closed and waiting to be uploaded to the NAS.
Synced	The order has been uploaded to the NAS. The local copy can be removed by cleaning up storage (section 3.7).

The list appears immediately even with thousands of files — order, type and date are taken directly from the file name. The number of records is counted in the background: until then a pulsing "…" is shown and fills in by itself. On the Remote (NAS) tab only the orders on the displayed page are counted; orders that have already passed through this machine show their count immediately.

3.2 Switches, date filter and sorting

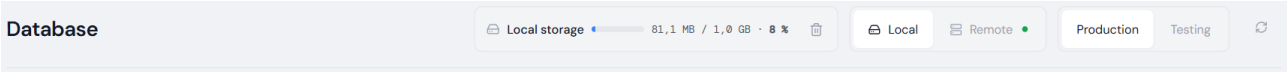


Figure 4 – Database header: storage indicator and switches

- **Local / Remote** — files on the machine PC, or on the NAS. A green dot next to "Remote" means the NAS is reachable; red means it is not.
- **Production / Testing** — production orders, or tests.
- **Circular arrows** — refresh the list manually (same as F5).
- **From / To** — limit the list to a period by creation date. The default is the last two days; **Clear filter** shows everything. If the filter hides all files, the application says how many are outside the period and offers a button to show the latest ones.
- **Sorting** — click a column header (Created, Order, Switch type, Records); click again to reverse the order.
- **Paging** — below the table. Set the number of files per page and the automatic refresh interval in Settings. New files in local storage appear immediately.

3.3 Expanding an order

Click the row of an order (or the arrow). A category distribution chart and a table of pieces with all parameters appear below the row. Click a piece to open its detail (chapter 5). The **Order detail** button at the bottom opens the whole order (chapter 4). For a test (Testing), clicking the row opens the test detail directly.

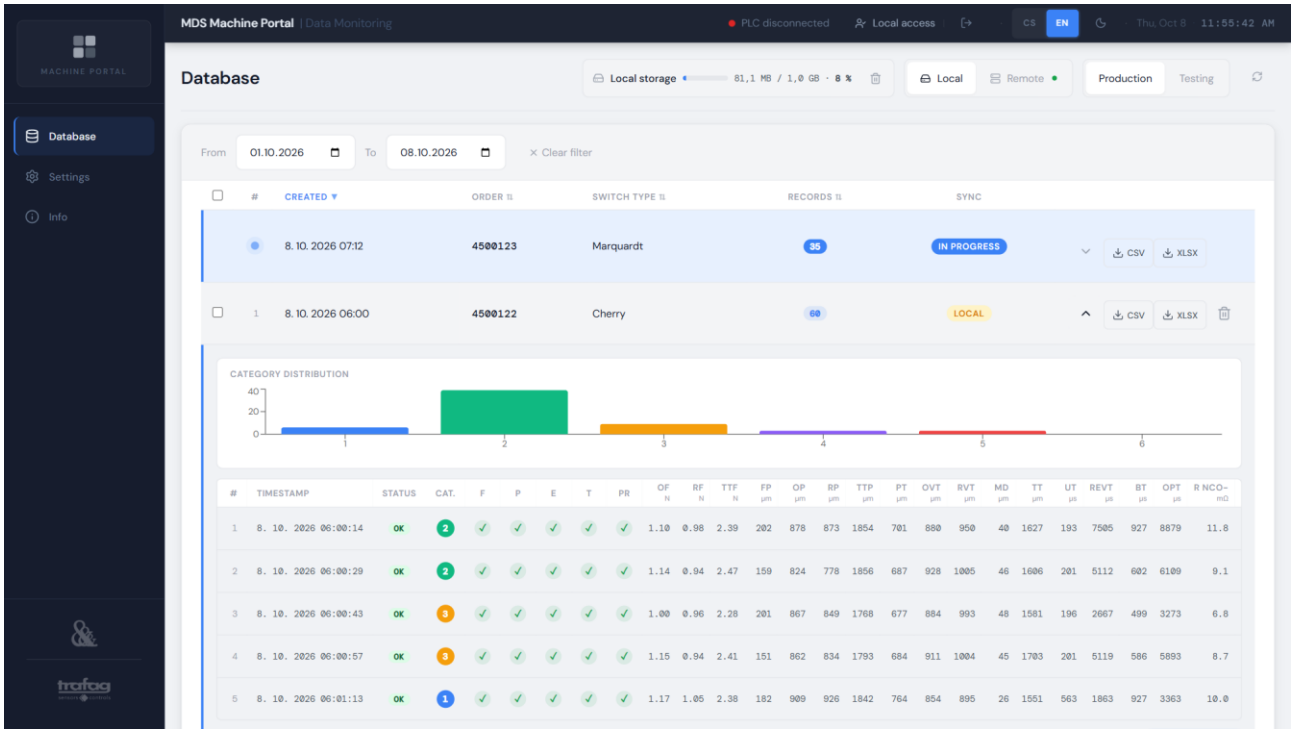


Figure 5 – Order expanded in the list

3.4 Downloading data

- **CSV** — the original file exactly as written by the line system (semicolon separator, UTF-8 encoding). Suitable for archiving and further processing.
- **XLSX** — all pieces of the order in one Excel sheet: numbers as numbers (you can calculate with them), time as a date, frozen header with a filter. The file is built by the server, so even an order with tens of thousands of pieces is ready in a few seconds. In Chrome and Edge a “Save as” dialog opens first after clicking (you can save directly to a USB drive or network folder), then the file is downloaded.

3.5 Deleting files

Only files in **local** storage can be deleted, only on the machine PC and with the Technician role or higher. Delete a single file with the bin icon; the application asks for confirmation. For batch deletion tick the files in the first column — a **Delete selected** / **Clear selection** bar appears above the table. At most 200 files can be deleted at once. An order in progress cannot be deleted.

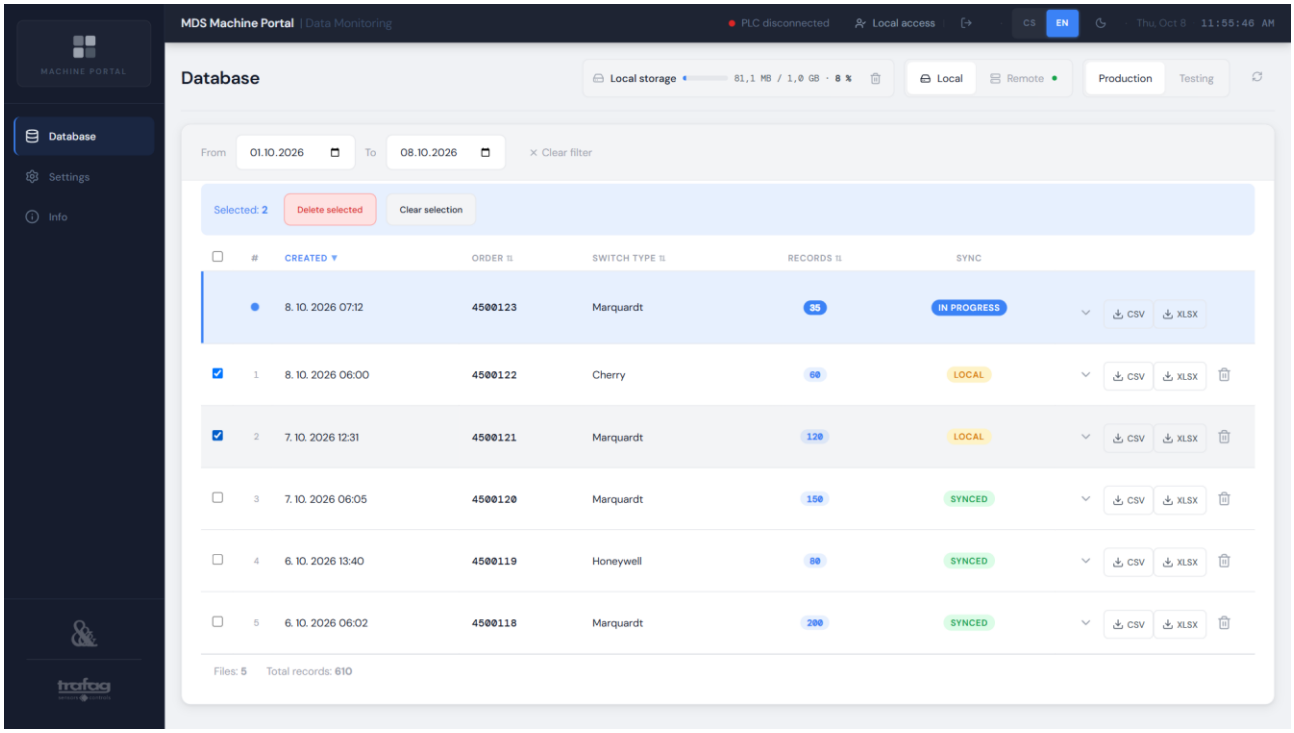


Figure 6 – Selecting several files for batch deletion

> **Caution:** Deletion cannot be undone. If the file is not yet on the NAS (Local badge), the order data is lost permanently.

3.6 Remote storage (NAS)

The **Remote** tab shows the archive of orders and tests on the NAS, including older ones no longer on the machine PC. It works the same way as local storage; files on the NAS cannot be deleted from the application. When the NAS is unreachable, a warning appears and the dot next to the tab is red — local data keeps working.

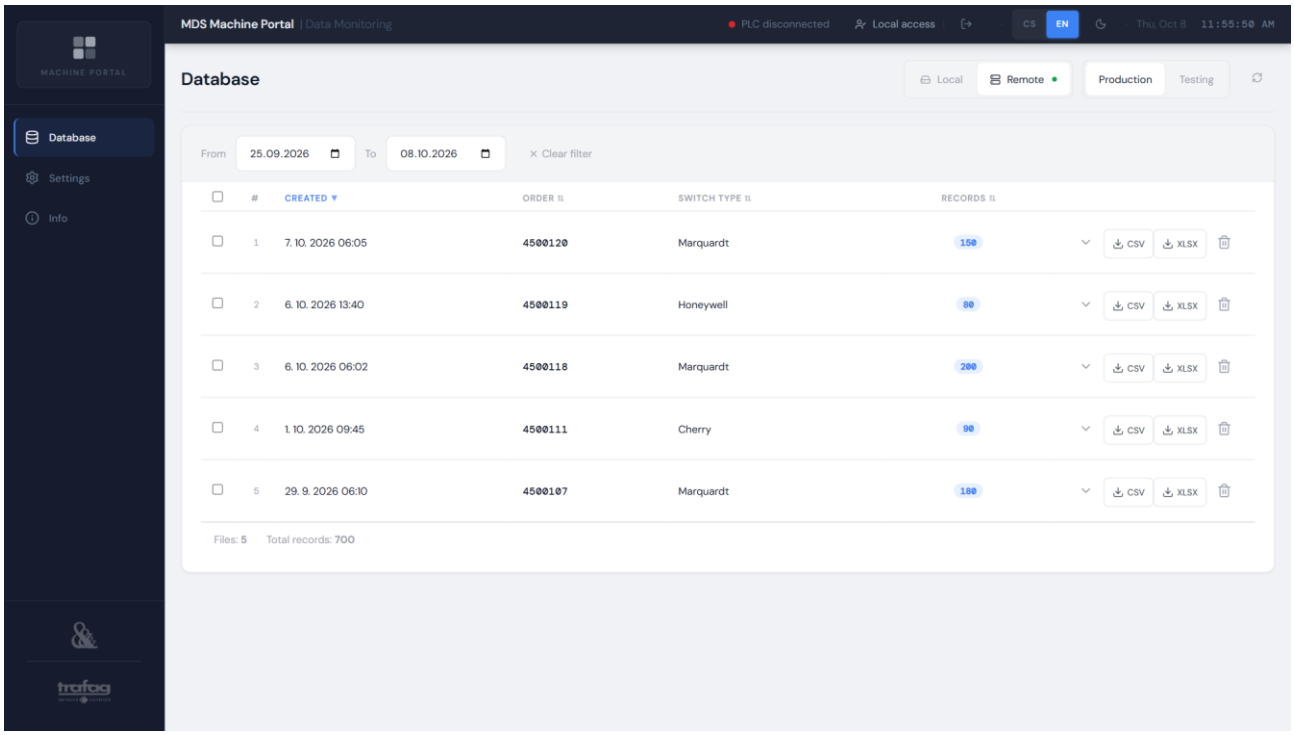


Figure 7 – Orders on the NAS

3.7 Local storage and clean-up

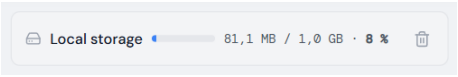


Figure 8 – Local storage indicator

The indicator in the Database header shows how much space the files on the machine PC take up compared to the configured limit. From 80 % it turns orange and also appears in the top bar; from 95 % it turns red. Free up space with the bin button **Clean up synced**:

1. Click the bin icon next to the indicator.
2. The dialog shows how many synced files will be deleted and their total size.
3. Confirm with **Clean up synced**. Every file is verified on the NAS (name and size) before deletion; anything missing or different on the NAS is kept.

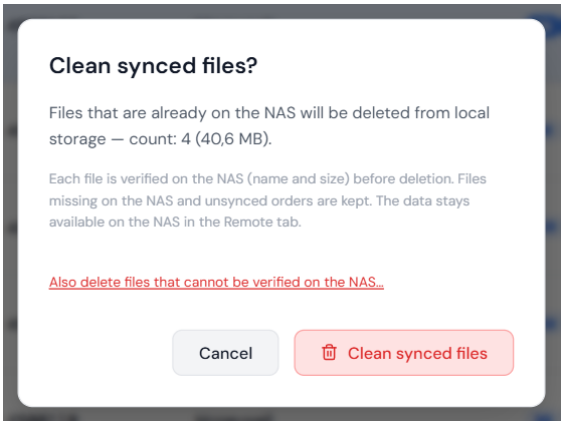


Figure 9 – Confirming clean-up of synced files

> **Caution:** The link “Also delete files that cannot be verified on the NAS…” deletes synced files without checking. Only use it if you know the data on the NAS is fine (e.g. the NAS is temporarily unreachable and the disk is full). The application asks you to confirm the risk.

3.8 Tests (Testing)

The **Testing** tab lists individual tests. Click a row (or the chart icon) to open the test detail (chapter 6).

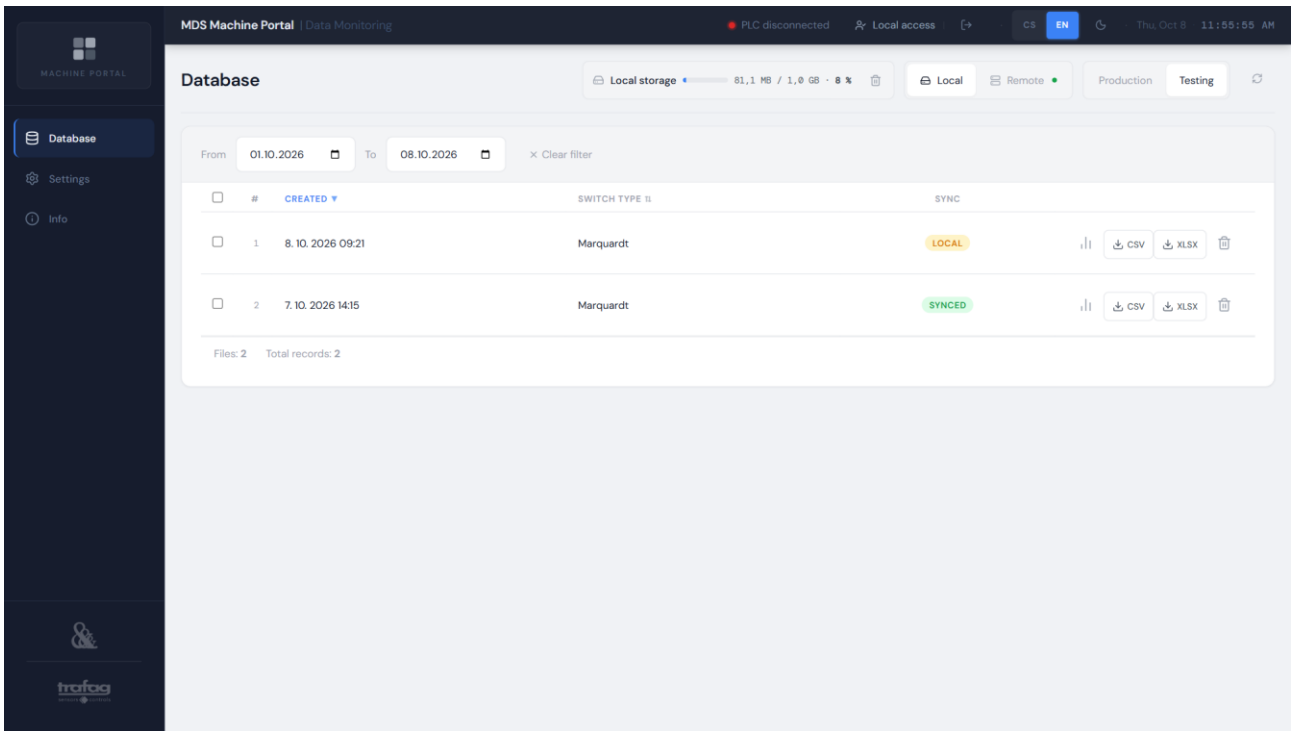


Figure 10 – Test list

4 Order detail

Open the order detail with the **Order detail** button in an expanded order. The **Back to database** button returns you to the same place in the list.

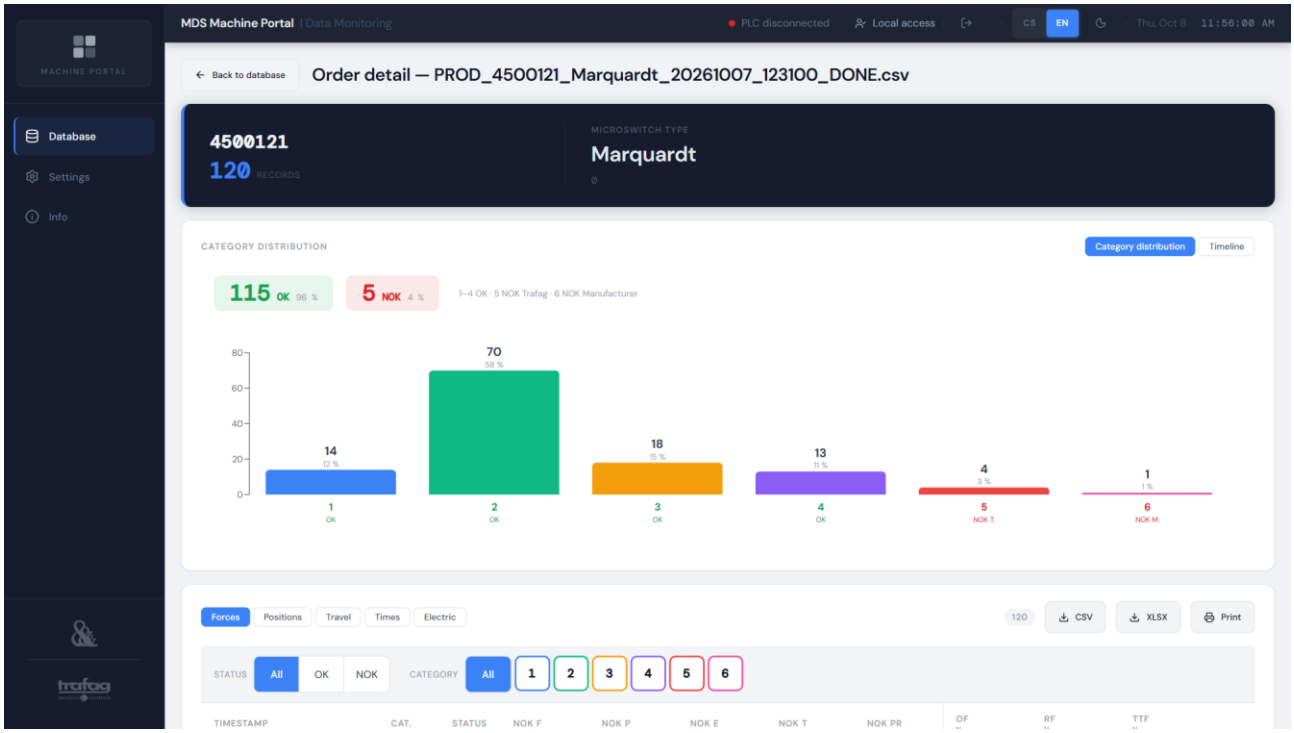


Figure 11 – Order detail — header and category distribution

4.1 Header and category distribution

The header shows the order number, the number of measured pieces and the microswitch type. Below it is a chart of how many pieces went into each category (sorting box), with an OK / NOK summary:

Table 5 – Sorting categories

Category	Meaning
1 – 4	OK — the piece passed; the number is the sorting box.
5	NOK Trafag — the piece failed Trafag requirements.
6	NOK manufacturer — the piece failed the switch manufacturer's requirements.

4.2 Timeline

Use the **Timeline** switch at the top right to see how production of the order progressed over time. The curve shows the number of pieces produced; flat sections are pauses.



Figure 12 – Order timeline

Table 6 – Timeline indicators

Value	Meaning
Start / End	Time of the first and last piece.
Total duration	From the first to the last piece, including pauses.
Average / median between pieces	Typical cycle time of one piece.
Longest pause	Longest gap between two pieces and when it occurred.
Pauses	Number and total of gaps longer than 5× the median and at least 1 minute.
Rate	Pieces per hour over the whole order duration (including pauses).

4.3 Piece table and filter

Below the chart is a table of all pieces. The **Forces**, **Positions**, **Travel**, **Times** and **Electric** tabs switch between parameter groups; time, category, status and NOK categories always stay visible. Click a column header to see a description of the parameter. A ✓ in a NOK category means OK, a red exclamation mark means a fault in that area (force, position, electric, times, process).

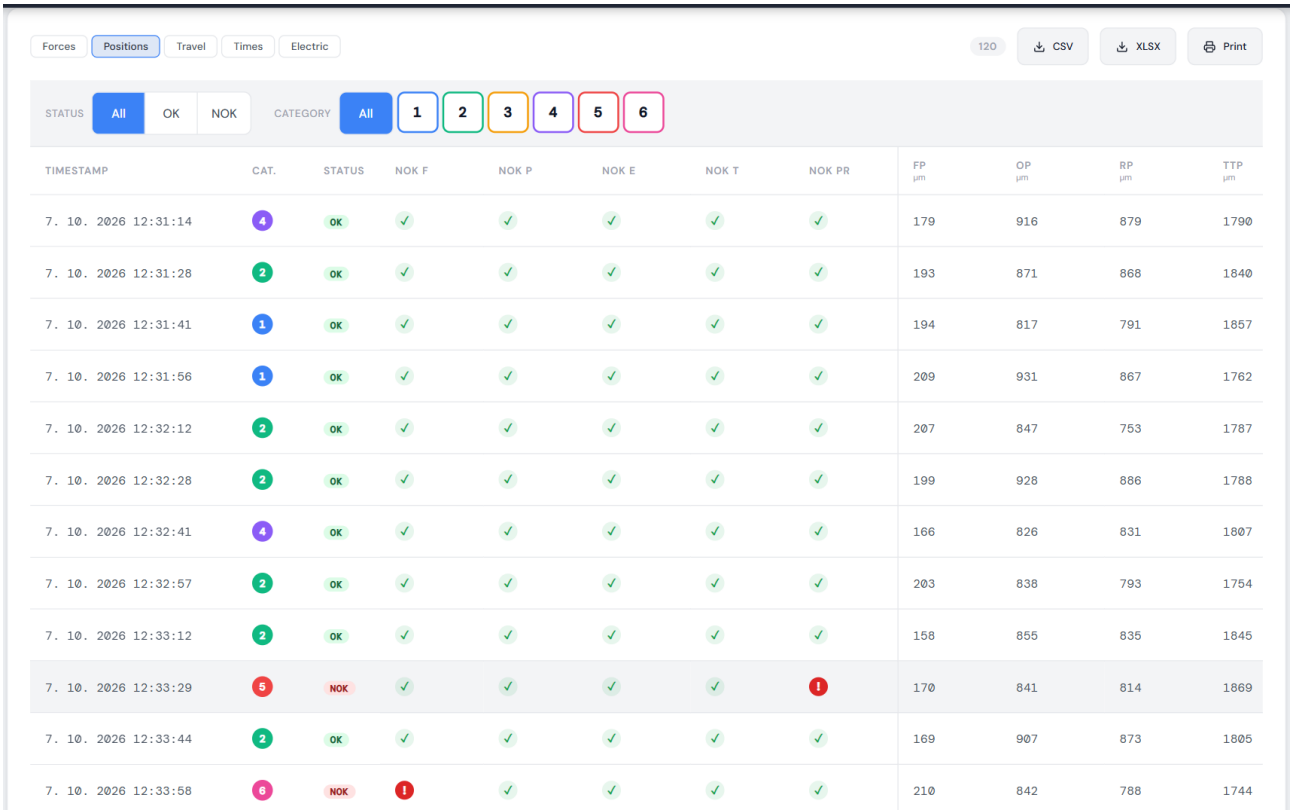


Figure 13 – Piece table — Positions group

The filter bar above the table narrows down the list of pieces:

- **Status** — All, OK or NOK.
- **Category** — All, or any combination of categories 1–6 (buttons in the box colours).
- Status and category apply together. With an active filter, “Shown: X of Y” and a **Clear filter** button appear.
- The category chart and the order header still show the whole order; the filter only changes the table.

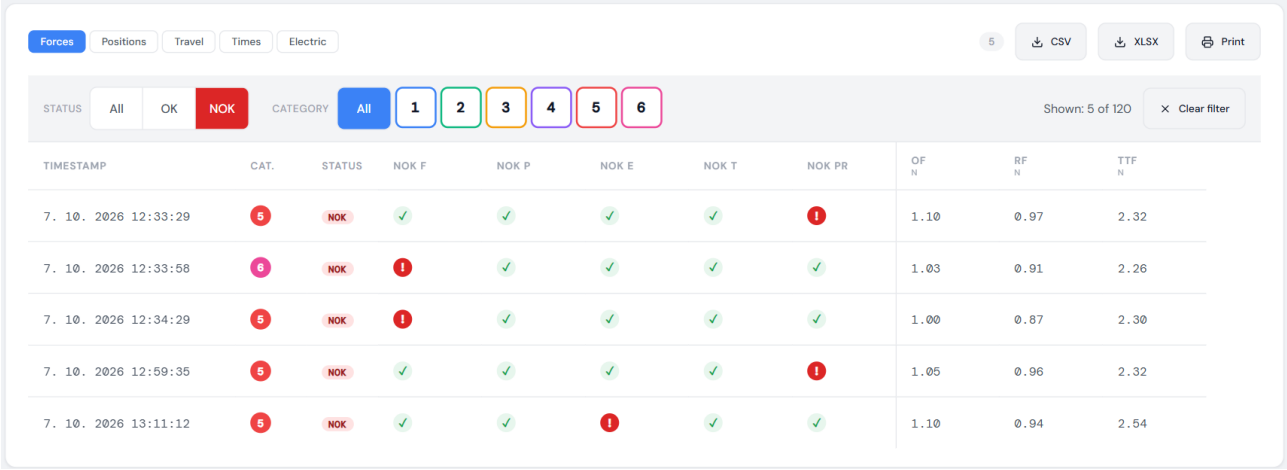


Figure 14 – Filter — NOK pieces only

4.4 Exporting and printing an order

The **CSV** and **XLSX** buttons above the table download the whole order (regardless of the filter). The **Print** button prepares an A4 report: order header, OK / NOK summary and tables of all parameter groups for the current page of the table (up to 200 pieces). For a larger order or with a filter, the top of the report states

which pieces it contains (e.g. “Printed pieces: 1–200 of 4,005 (page 1 of 21)”) and which filter applies; the # column is the position of the piece in the whole order. Download a whole large order as CSV or XLSX. Charts are not included in the order report. The report header shows the print time and the signed-in user; the footer shows the page number.

5 Record detail (one piece)

Open the detail of a piece by clicking its row in the piece table. The heading shows the position of the piece in the order (e.g. 4 / 120); the summary line shows the order, type, measurement time, category and OK / NOK status.

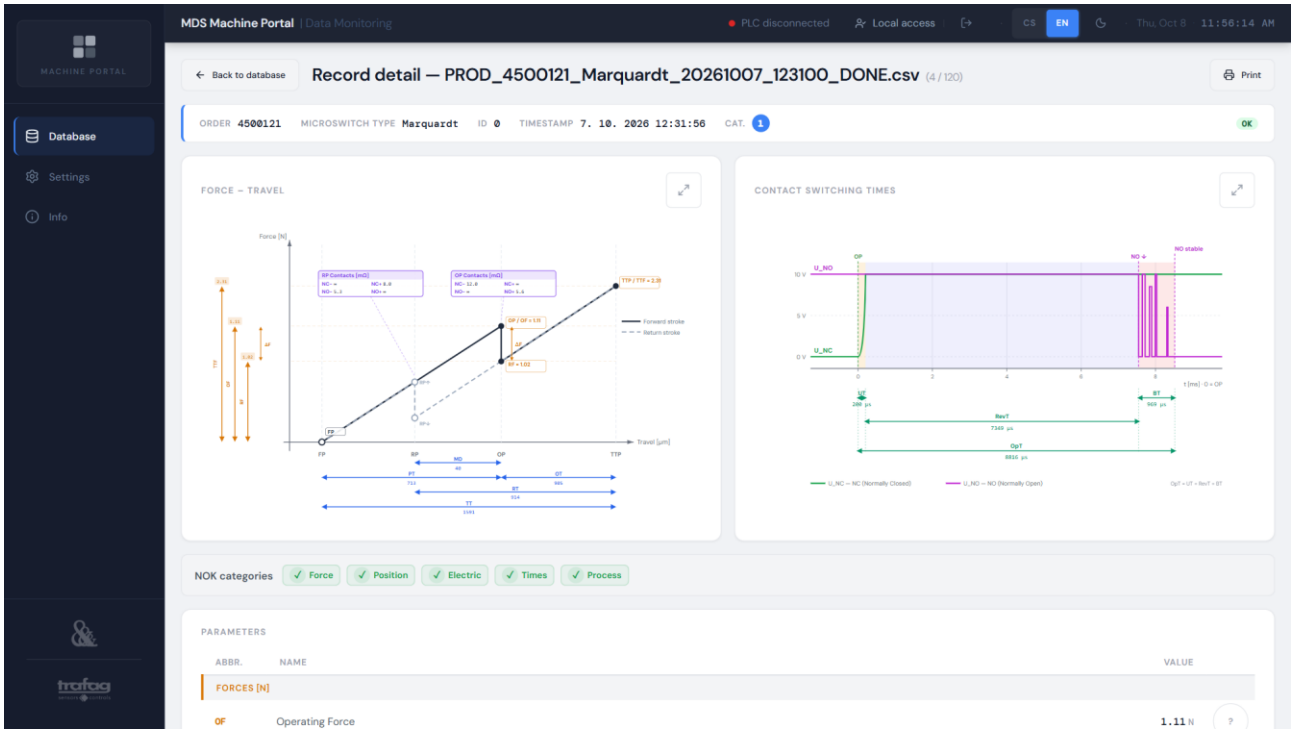


Figure 15 – Record detail — diagrams

- **Force – travel** — force during pressing (solid line) and release (dashed) with the FP, OP, RP and TTP points, the OF, RF and TTF forces, the PT, OT, RT, MD and TT travels and the contact resistances at OP and RP.
- **Contact switching times** — NC and NO contact voltage around the switching moment, to time scale, with the UT, RevT, BT and OpT times.
- Use the arrows button in the corner of a diagram to enlarge it to full screen.
- Below the diagrams are the NOK categories and a table of all parameters. The **?** button next to a parameter shows its explanation.

The **Print** button creates an A4 report with both diagrams and the parameter table:

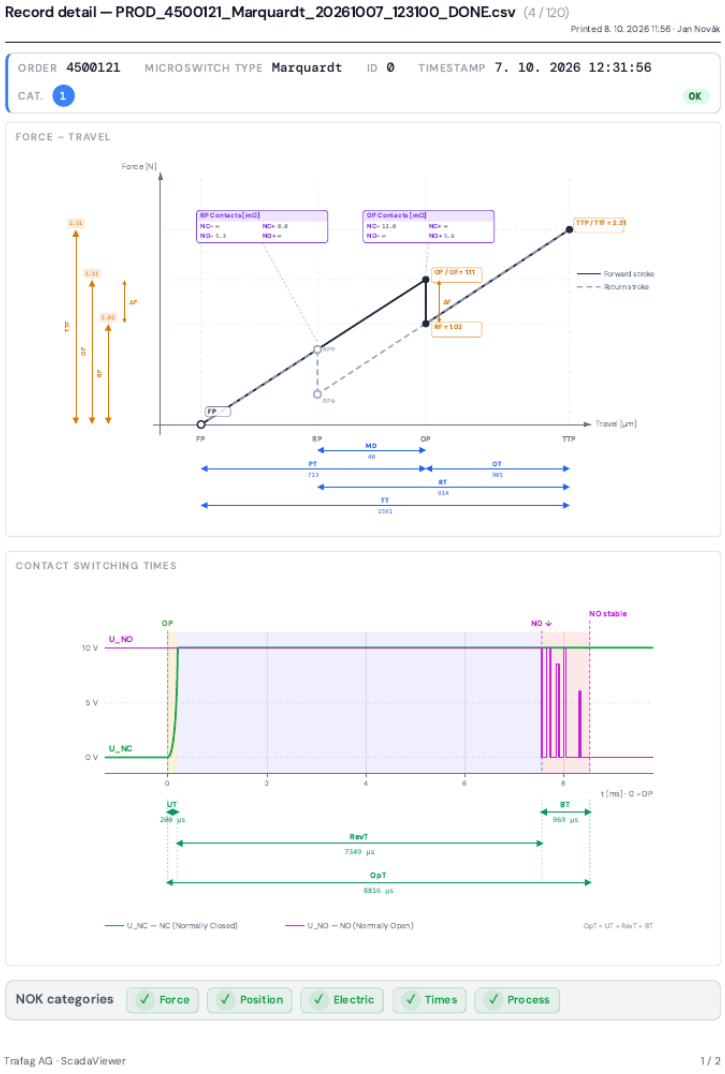


Figure 16 – Printed record report (page 1)

> **Tip:** All parameters and their abbreviations are described in Appendix A.

6 Test detail (Testing)

A test is a detailed measurement of one microswitch including the course of all signals. The header shows the switch type and ID, measurement time and duration and the overall OK / NOK result. The content is divided into tabs.

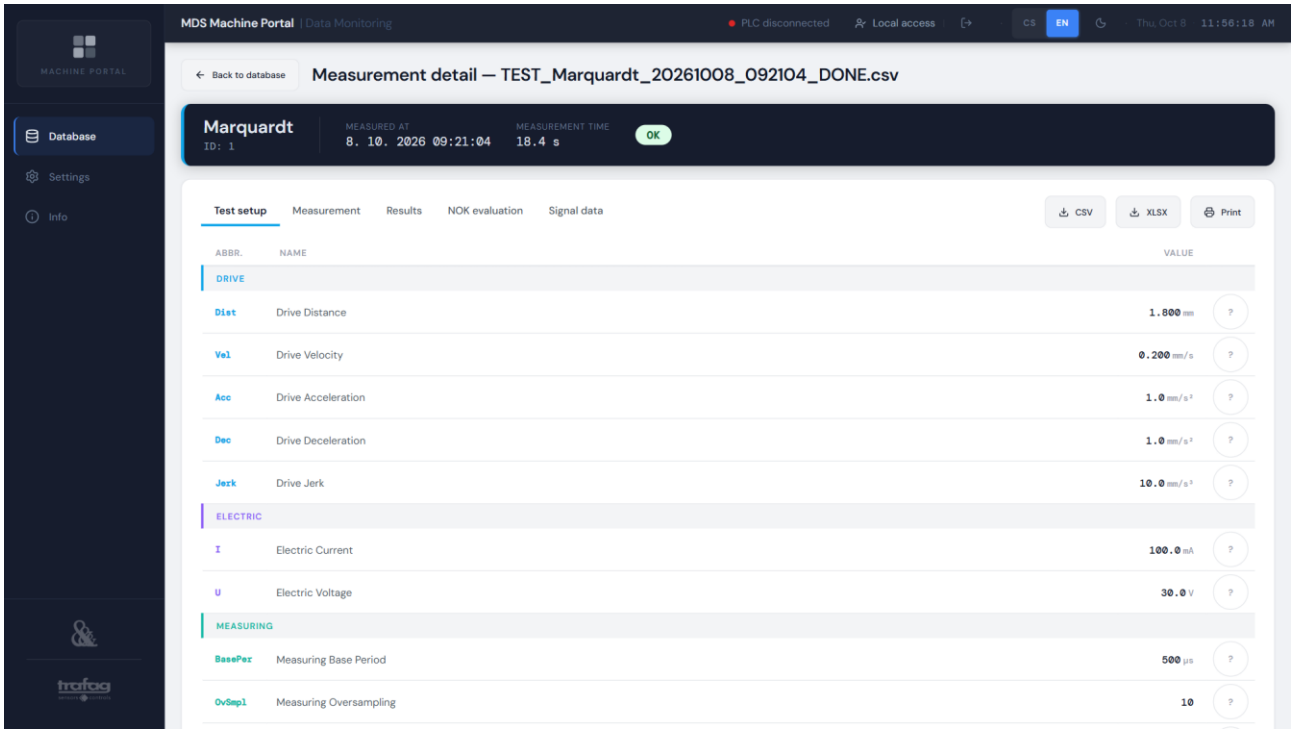


Figure 17 – Test detail — Test settings

Table 7 – Test detail tabs

Tab	Content
Test settings	Test inputs: drive distance, velocity, acceleration and jerk, measuring current and voltage, sampling (period in µs, oversampling, samples reserve) and safety limits for distance and force.
Measurement	Measurement duration and the time recorded by the controller.
Results	Calculated parameters: forces, positions, travels, switching times and contact resistances.
NOK evaluation	Overall result and the status of the five NOK categories (Force, Position, Electric, Times, Process).
Signal data	Charts of the measured signals — see 6.1.

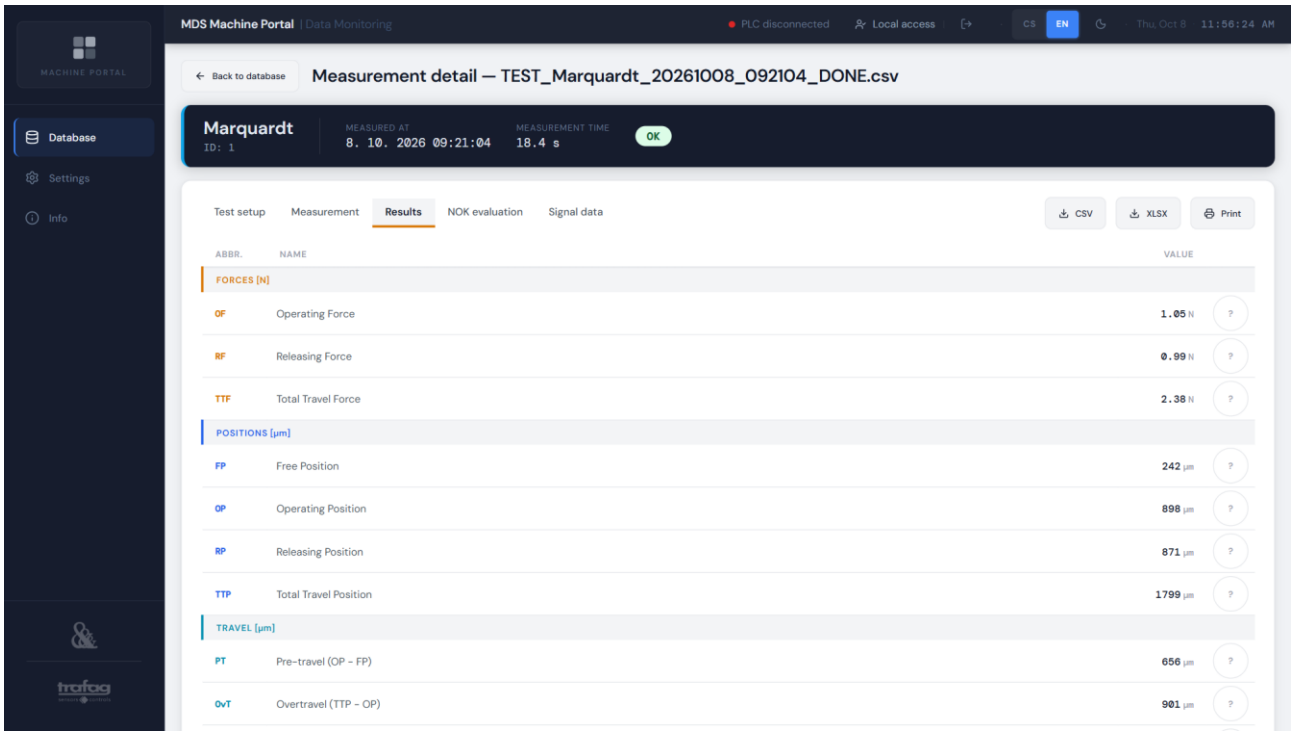


Figure 18 – Test results

6.1 Signal data

The **Signal data** tab shows the measurement in charts. The number of samples is shown at the top right. The charts have five sub-tabs:

Table 8 – Signal data sub-tabs

Sub-tab	What it shows
Overview	Position, force, contact voltage, current and resistance over time for the whole measurement.
Results	Position and force with the FP, OP, TTP, RP points marked, plus a table of analysis results.
Hysteresis	Force versus position for the forward and reverse stroke.
Switching detail	Voltage, current and force around the OP and RP points at high resolution.
Timing	Contact voltage and current with the UT, RevT, BT and OpT times marked.

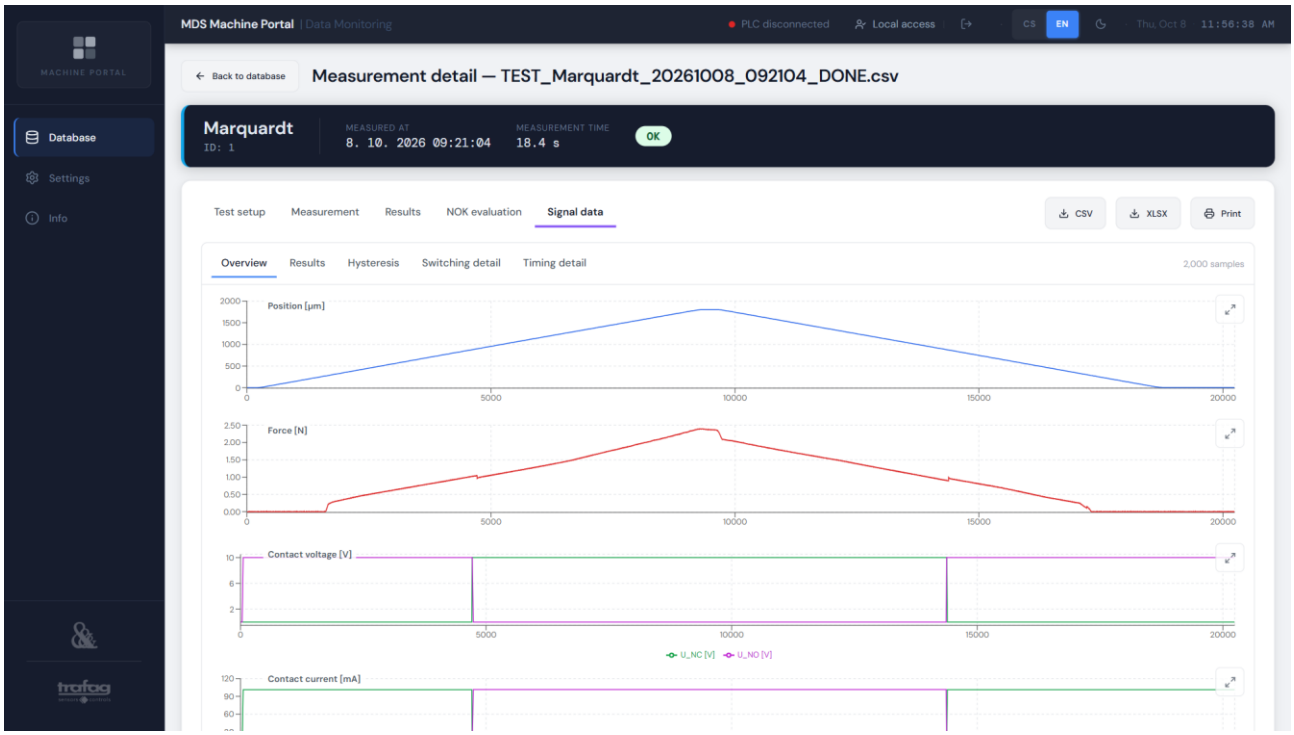


Figure 19 – Signal data — Overview

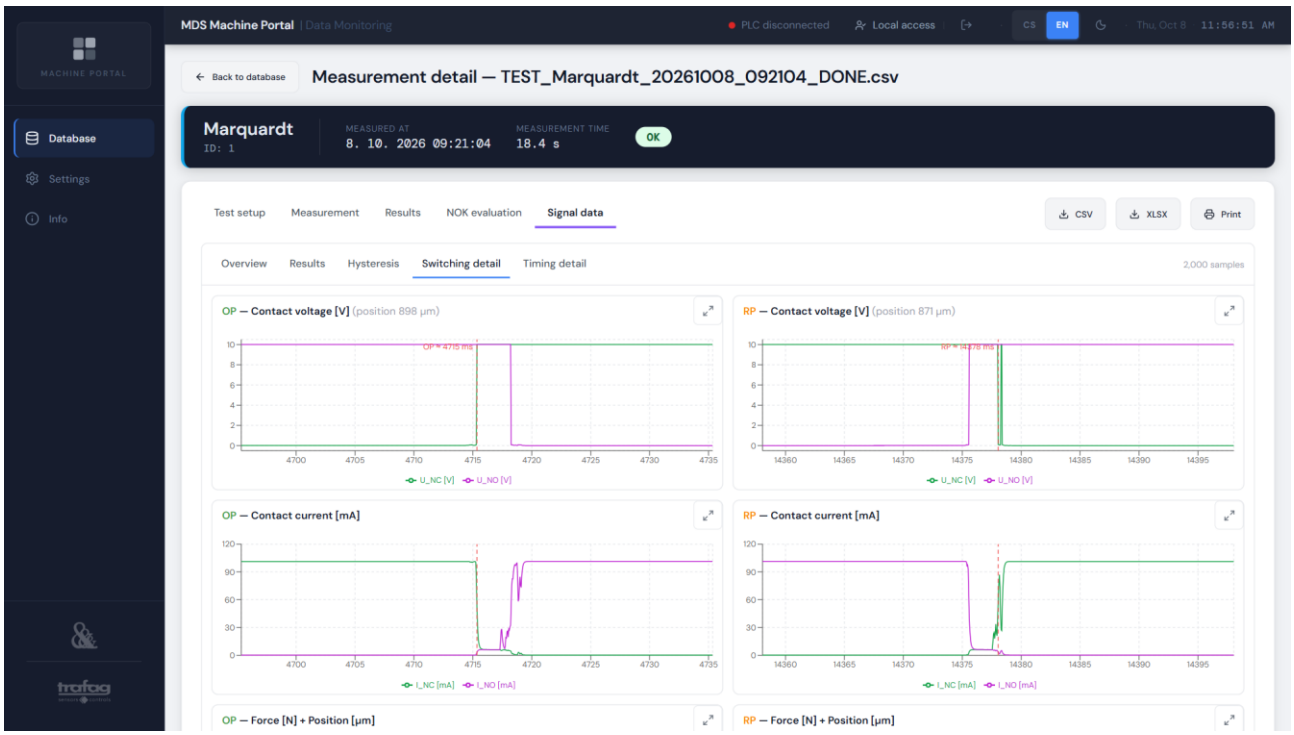


Figure 20 – Signal data — Switching detail

Zooming a chart

- **Ctrl + mouse wheel** or **two fingers** on a touch screen — zoom the time axis. The zoomed section is loaded at full resolution.
- **Drag** — move the zoomed section.
- **Double-click** — return to the default size.
- **Arrows button** in the corner of a chart — full screen; close with **Esc**.

6.2 Printing a test report

The **Print** button (or **Ctrl + P**) prepares a report with all tabs — settings, measurement, results, NOK evaluation — and all signal data charts. Preparing the charts takes a few seconds (the button shows “Preparing print…”), then the browser print dialog opens.

7 Settings

7.1 Preferences

Language, dark / light appearance, number of files per Database page and the automatic list refresh interval. Preferences are stored in the browser, so each computer can have its own. The ⓘ icon next to a row shows help.

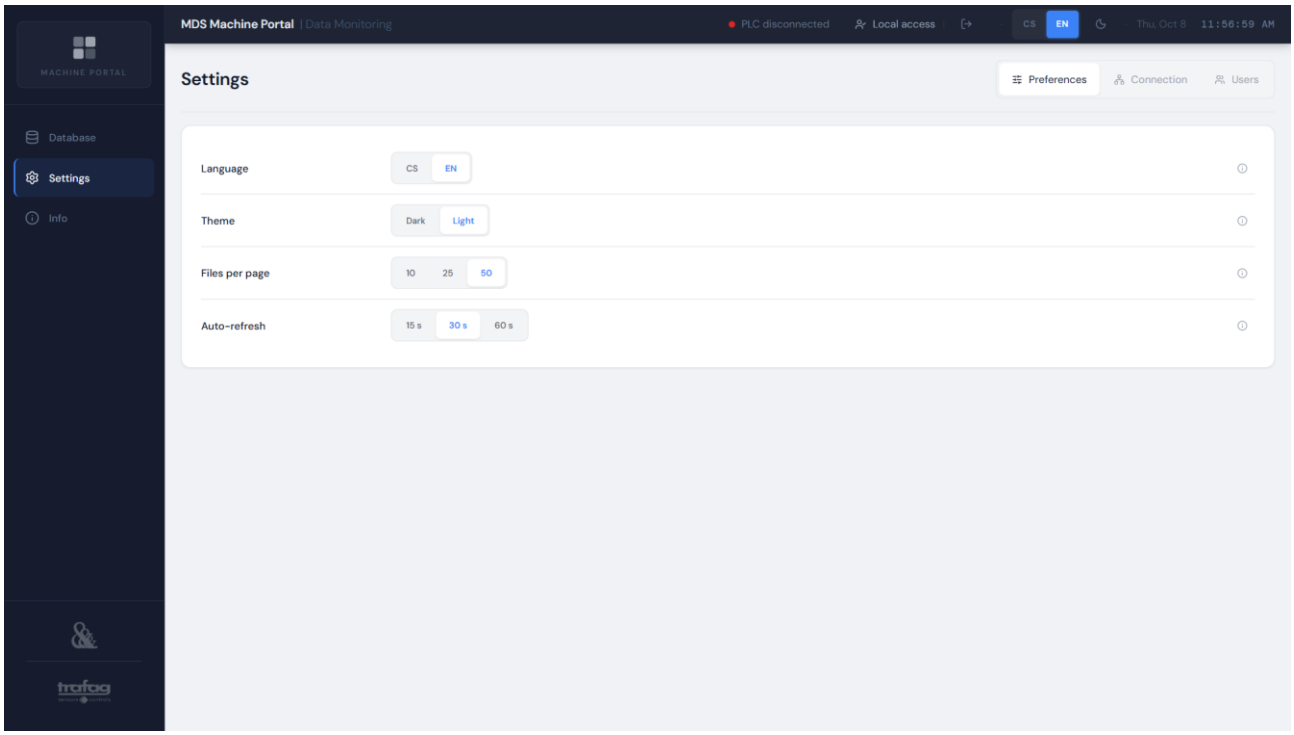


Figure 21 – Settings — Preferences

7.2 Connection

PLC connection status (Net ID, port), availability of local storage and the NAS with their paths, and usage and limit of local storage. Only an administrator can change the paths and the limit, and only on the machine PC; changes take effect immediately. The folder button lets you pick the local path by browsing the disk.

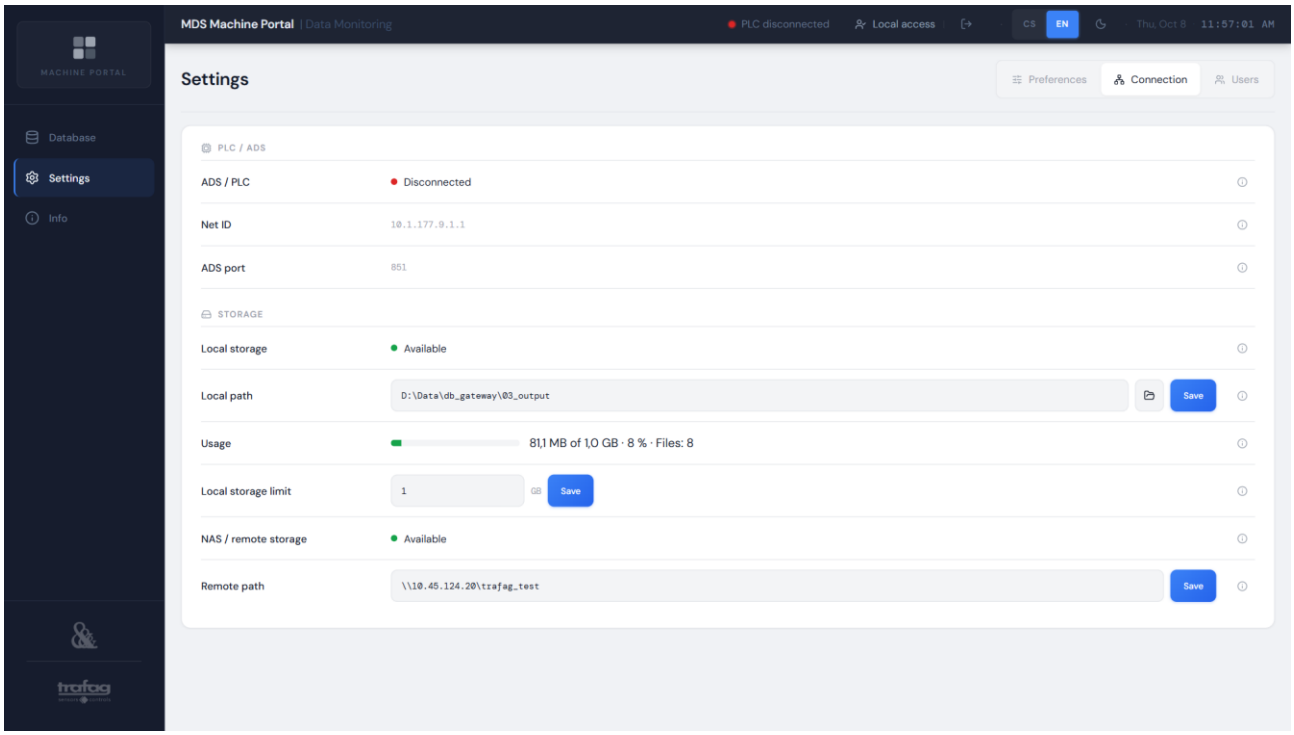


Figure 22 – Settings — Connection

7.3 Users

This tab is visible only to administrators and the manufacturer. It lists the accounts and lets you add a user, change a password and delete an account. An administrator cannot create, delete or change the password of an account with the same or a higher role. Every user can change their own password.

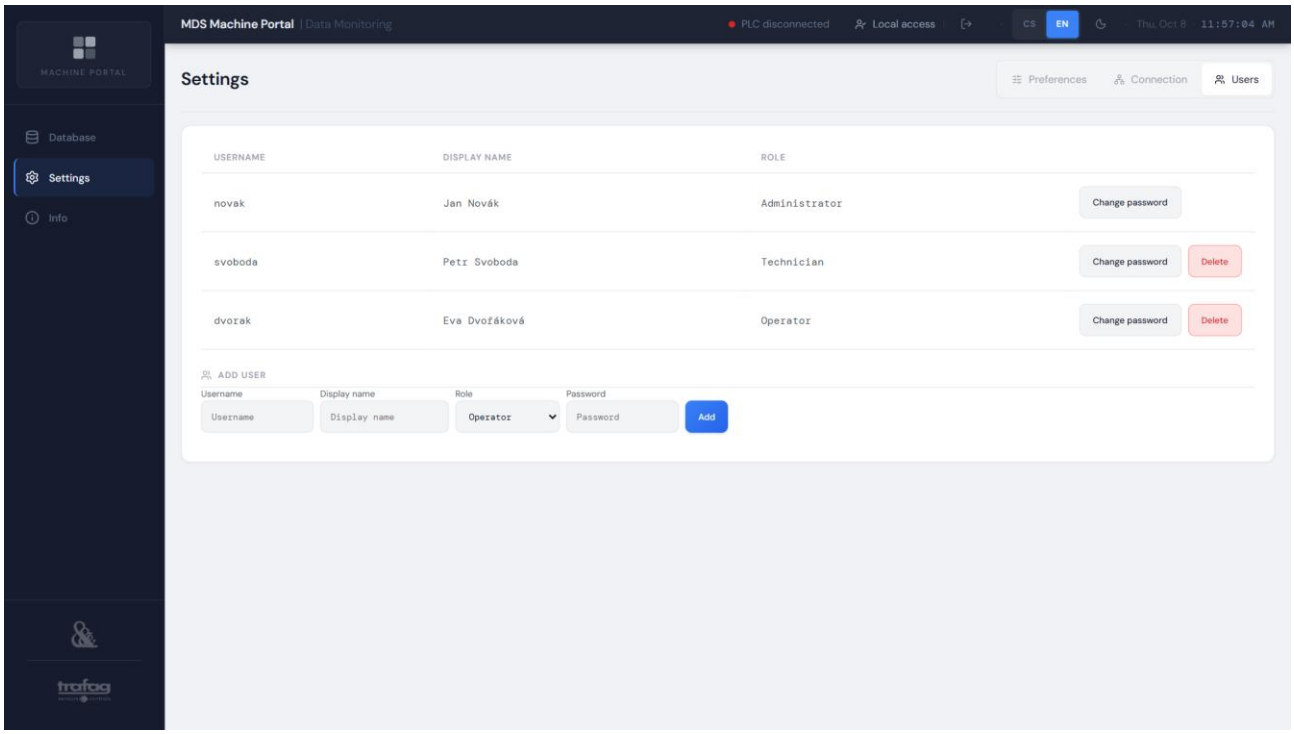


Figure 23 – Settings — Users

8 Info

The Info page shows the application version, project number, customer, supplier and contact. Please quote the version when reporting a problem.

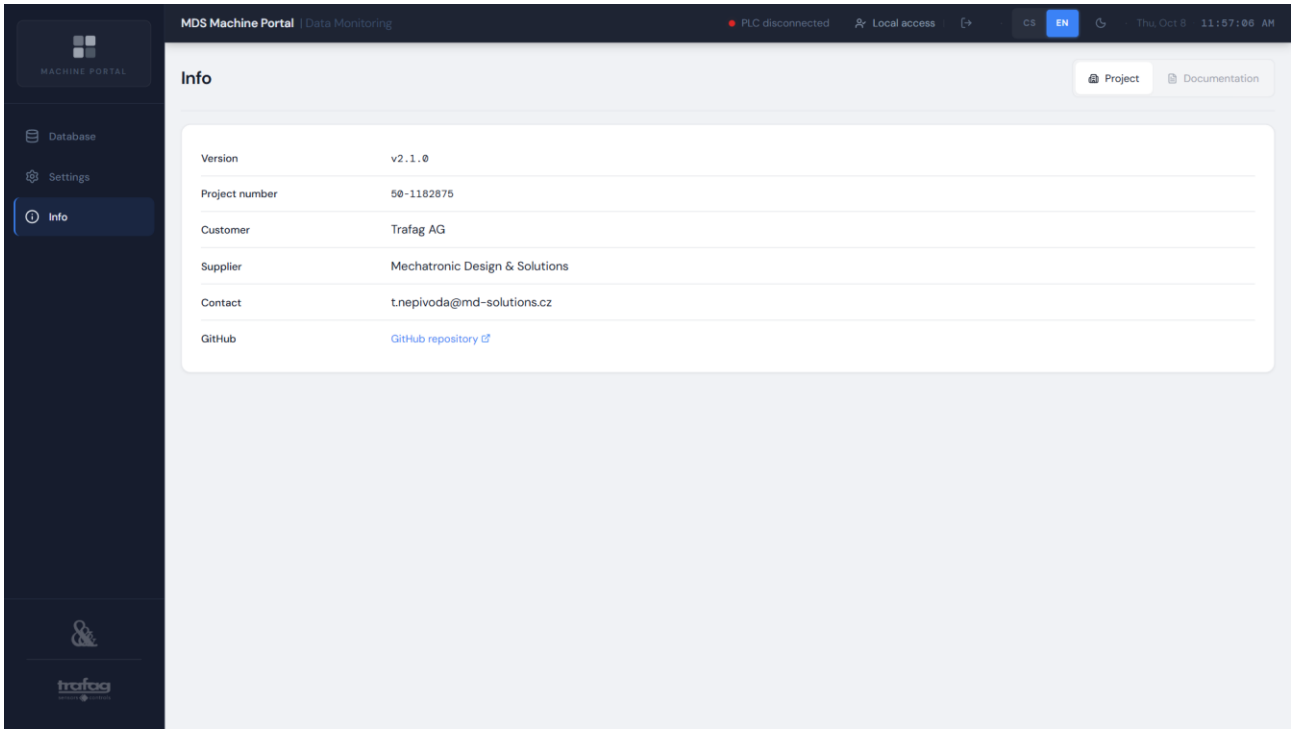


Figure 24 – Info page

9 Common situations

Table 9 – Common situations and how to resolve them

Situation	Cause and solution
“PLC disconnected” in the top bar	The machine controller is unreachable (switched off, restarting, cable unplugged). Database data can still be viewed. The connection is restored automatically.
Red bar “Server unavailable”	The browser lost the connection to the application server (PC or service restart, network outage). Wait, the bar disappears by itself. If it persists, contact your administrator.
The Remote tab is unavailable	The NAS cannot be reached on the network. Local data keeps working and orders will be uploaded once the NAS is reachable.
An order is missing in the Database	Check the From / To filter (default: last two days) and the Local / Remote and Production / Testing switches.
“…” instead of the number of records	The count is being calculated in the background and fills in by itself (for large files on the NAS depending on network speed). If it takes long, check that the NAS is reachable.
“Session expired”	A sign-in is valid for 8 hours. Sign in again.
Sign-in is blocked	After 5 wrong passwords, sign-in from that computer is blocked for 10 minutes. Wait and try again.
Storage filling up warning	Free up space with Clean up synced (section 3.7). If only Local files remain, check the NAS connection.
Cannot delete or change settings	You are connected from another computer (Remote access label), or you do not have the required role (Table 1).

Appendix A Parameter overview

Parameter abbreviations and names are shown in English in the application (technical names as used by the customer). Values: forces in N (2 decimal places), positions and travels in µm, times in µs, contact resistances in mΩ; an open contact (≥ 1 MΩ) is shown as ∞.

Table 10 – Forces [N]

Abbr.	Name	Description
OF	Operating Force	Force [N] at the operating point (OP). If it is too high or too low, the switch does not meet the specification.
RF	Releasing Force	Force [N] at which the contact releases on the return stroke (RP). Together with OF it defines the force hysteresis.
TTF	Total Travel Force	Maximum force [N] at the end of travel (TTP). Must not exceed the permitted limit for the switch type.

Table 11 – Positions [µm]

Abbr.	Name	Description
FP	Free Position	Rest position of the actuator [µm] without external force — the reference point for all travel values.
OP	Operating Position	Position [µm] where the contact switches on the forward stroke (absolute position). PT = OP − FP.
RP	Releasing Position	Position [µm] where the contact releases on the return stroke (absolute position). MD = OP − RP.
TTP	Total Travel Position	Position [µm] at the end of travel — the deepest point of actuation during the measurement. TT = TTP − FP.

Table 12 – Travel [µm]

Abbr.	Name	Description
PT	Pre-travel [µm] (OP − FP)	Pre-travel [µm] — distance from FP to OP (PT = OP − FP). Must be sufficient for reliable switching.
OvT	Overtravel [µm] (TTP − OP)	Overtravel [µm] — reserve beyond the operating point (OT = TTP − OP). Protects the contact from overload.
RvT	Releasing Travel [µm] (TTP − RP)	Releasing travel [µm] — distance from the end of travel back to the releasing point (RT = TTP − RP).
MD	Movement Differential [µm] (OP − RP)	Movement differential [µm] — distance between OP and RP (MD = OP − RP, position hysteresis). Larger MD = more stable switching. Negative MD = negative hysteresis (RP lies beyond OP) — abnormal switch behaviour.
TT	Total Travel [µm] (TTP − FP)	Total travel [µm] — distance from FP to TTP (TT = TTP − FP).

Table 13 – Times [µs]

Abbr.	Name	Description
UT	Unstable Time	Unstable time [µs] — duration of contact oscillation right after switching, before the contact settles.
RevT	Reverse Time	Reverse time [µs] — window in which the contact briefly reverses towards the closed state (NC closes for a moment).
BT	Bounce Time	Bounce time [µs] — total duration of contact bounce after switching. Longer BT = more noise, slower response.
OpT	Operating Time [µs] (UT + RevT + BT)	Operating time [µs] — total switching time, the sum UT + RevT + BT.

Table 14 – Electric [mΩ]

Abbr.	Name	Description
R NCo−	NC	Resistance [mΩ] of the normally closed (NC) contact at the operating point (OP), negative pole. High resistance = degradation.
R NCo+	NC	Resistance [mΩ] of the NC contact at the operating point (OP), positive pole.
R NCr−	NC	Resistance [mΩ] of the NC contact at the releasing point (RP), negative pole.
R NCr+	NC	Resistance [mΩ] of the NC contact at the releasing point (RP), positive pole.
R NOo−	NO	Resistance [mΩ] of the normally open (NO) contact at the operating point (OP), negative pole. Low resistance = properly closed.
R NOo+	NO	Resistance [mΩ] of the NO contact at the operating point (OP), positive pole.
R NOr−	NO	Resistance [mΩ] of the NO contact at the releasing point (RP), negative pole.
R NOr+	NO	Resistance [mΩ] of the NO contact at the releasing point (RP), positive pole.